

Matter, with its evidence.

Materials Atlas Table Codex

Representative publishing proof

A five-page sample of the local book: reading guidance, a precision table, decay equations, evaluated nuclear states and the coverage boundary for a developing element record.

This PDF is a separate typeset proof of selected source material. The complete working HTML edition remains the full book. It is not a browser-generated PDF of every chapter.

Read in context

Collection	Meaning
MAT Core Data	Properties and evaluations retain units, conditions, uncertainty and source locators.
Research / Emerging	Imported claims and developing results retain their pending review status.
Historical / Alternative / Claims	Original author and historical material remain separate from measured scientific properties.

Reading path

Start with an element identity. Follow its detailed tables and structured data, inspect the source and evidence status, then explore related records and visuals. Unknown and planned fields remain visible.

Canonical source: docs/00-front-matter/03-How-to-Use-MAT-Codex.md

Hydrogen

Atomic identity & ground-state properties

Selected precision values are reproduced from the existing Hydrogen table. Conditions and bibliography aliases travel with the values; this proof adds no new scientific validation.

Property	Value	Unit	Conditions / State	Source
Atomic number	1	—	elemental identity	SRC-H-001
Proton count	1	proton	all Hydrogen nuclei	—
Neutral electron count	1	electron	neutral H atom	—
Ground configuration	1s ¹	—	neutral atomic H	SRC-H-001
Ground level	² S _{1/2}	—	neutral atomic H	SRC-H-001
First ionization energy	13.598434599702 ± 0.000000000012	eV	ground-state neutral atom	SRC-H-001
Electron affinity	0.754195 ± 0.000018	eV	selected precision measurement	SRC-H-003
Standard atomic weight	[1.00784, 1.00811]	—	normal terrestrial materials	SRC-H-002

Canonical source: records/0001-Hydrogen-H/tables/0001-Hydrogen-H-TABLE-002-Atomic-Identity.md

Source visibility

SRC-H-001, SRC-H-002 and SRC-H-003 are preserved local bibliography identifiers. Consult the Hydrogen source registry for the original references and their scope. Empty source cells remain visible as inherited gaps.

Canonical source: records/0001-Hydrogen-H/sources/ and data/registries/sources.yaml

Sodium-24 decay

Calculation 003 · beta-minus Q value

The following equations and atomic masses are retained from supplied 42.txt, line 987. Their original validation assertions have not been adopted as independent test results.



$$M(^{24}\text{Na}) = 23.990962950 \text{ u}$$

$$M(^{24}\text{Mg}) = 23.985041697 \text{ u}$$

$$Q_{\beta-} = [M_p - M_d]c^2$$

$$Q_{\beta-} \approx 5.51561 \text{ MeV}$$

Interpretation

This total Q value must not be confused with the electron endpoint for a particular branch. The source notes that much of the decay proceeds through excited magnesium-24 levels followed by gamma emission.

Canonical source: records/0011-Sodium-Na/calculations/0011-Sodium-Na-CALC-003-Na24-BetaMinus-Q.md

Canonical source: archive/source-intake/mat-42-52/42.txt · line 987

Mass uncertainty propagation and the supporting evaluation for these input masses still require claim-level review.

Evaluated nuclear states

A traceable excerpt from NUBASE2020

These ground-state rows are selected from the retained evaluation. The full element dataset also keeps separately indexed excited states, inequalities, estimated values and the original source lines.

Nuclide	N	Mass excess / keV	Half-life	Source line
24Mg	12	-13933.578 ± 0.013 keV	stbl	226
25Mg	13	-13192.78 ± 0.05 keV	stbl	239
26Mg	14	-16214.544 ± 0.029 keV	stbl	251
27Mg	15	-14586.59 ± 0.05 keV	9.435 ± 0.027 m	263
28Mg	16	-15019.95 ± 0.26 keV	20.915 ± 0.009 h	275

What “stable” means here

The literal source token “stbl” is kept. The structured value is null with NOT_APPLICABLE; it is not replaced by a guessed numerical half-life. A # elsewhere in the evaluation denotes an estimate from systematics.

Canonical source: records/0012-Magnesium-Mg/data/isotopes/0012-Magnesium-Mg-NUBASE2020-Evaluation.yaml

F. G. Kondev et al., The NUBASE2020 evaluation of nuclear physics properties, published 2021. doi:10.1088/1674-1137/abddae · SRC-000008.

Snapshot SHA-256: 1585a5eea86c5e17e90307c7e6e786d060049c4039e392a261ff6db977df9859

Evaluation year 2020; retrieved 8 September 2026. Later measurements need a separate accession and review.

Make the boundary visible.

Aluminium has a selectable baseline identity, an imported research expansion and 32 evaluated nuclear states. Its main record remains BASELINE.

Available	Review still needed
Identity and direct element navigation	Full prose, engineering conditions and remaining material properties
Structured nuclear evaluation and source lines	Later research, field-level scientific review and complete uncertainty coverage
Supplied research, citations and claims ledger	Independent verification of the imported scientific assertions
Existing figures and explicit placeholders	Final figure descriptions, rights and publication review

Coverage across the book

The generated matrix covers all 119 record packages, including Origin State. The retained nuclear evaluation covers all 118 chemical elements. File counts and passing software checks do not measure scientific completeness.

Canonical source: docs/05-index/Record-Coverage.md

Canonical source: data/quality/record-coverage.json

Latest review intake

107 numbered recommendations from two supplied reviews are preserved with their original text and source locators. Domain vocabularies and reading depths remain proposals until there is reviewed content to support them.

Canonical source: docs/09-production/04-Review-Brief-Reconciliation.md